Transactions of the Royal Society of Tropical Medicine and Hygiene (2004) **98**, 142–147





www.elsevierhealth.com/journals/trst

Evaluation of a species-specific PCR assay for the *Anopheles funestus* group from eleven African countries and Madagascar

M.M. Weeto[a,*], L.L. Koekemoer[a], L. Kamau[b], R.H. Hunt[c], M. Coetzee[a]

[a] *Vector Control Reference Unit, National Institute for Communicable Diseases and Department of Clinical Microbiology and Infectious Diseases, School of Pathology, University of the Witwatersrand, P.O. Box 1038, Johannesburg 2000, South Africa*
[b] *Centre for Biotechnology Research and Development, Kenya Medical Research Institute, P.O. Box 54840, Nairobi, Kenya*
[c] *Department of Animal, Plant and Environmental Sciences, University of the Witwatersrand, Johannesburg, South Africa*

Received 3 March 2003; received in revised form 27 May 2003; accepted 2 June 2003

KEYWORDS
Anopheles funestus;
An. parensis;
An. vaneedeni;
An. rivulorum;
An. leesoni;
PCR;
Africa

Summary A newly published cocktail polymerase chain reaction (PCR) assay can identify five members of the *Anopheles funestus* group: *An. funestus*, *An. vaneedeni*, *An. parensis*, *An. leesoni* and *An. rivulorum*. The assay was evaluated on specimens from 11 African countries: Angola, Cote d'Ivoire, Ethiopia, Kenya, Malawi, Mozambique, Namibia, South Africa, Tanzania, Uganda and Zambia; and the island of Madagascar. The polymerase chain reaction single strand conformation polymorphism (PCR-SSCP) and the internal transcribed spacer PCR (ITS2-PCR) assays were used as *a priori* identification methods on 900 specimens. Of these, 96.4% were correctly identified using the new cocktail PCR method. The remaining 3.5% (32) from Malawi failed to amplify. The failure to identify these samples was not significantly different from the samples from the other countries (Chi-square; $P > 0.05$). The results suggest that the new species-specific PCR assay is an efficient and effective means of identifying 5 members of the *An. funestus* group in a single reaction. It is also less time-consuming than the molecular methods previously used on the group.
© 2003 Royal Society of Tropical Medicine and Hygiene. Published by Elsevier Ltd. All rights reserved.

1. Introduction

The *Anopheles funestus* group consists of 9 species that are widely distributed throughout subtropical Africa (Gillies and Coetzee, 1987; Gillies and De Meillon, 1968). Members of the group include *An. funestus* Giles, *An. vaneedeni* Gillies and Coetzee, *An. leesoni* Evans, *An. parensis* Gillies, *An. rivulorum* Leeson, *An. fuscivenosus* Leeson, *An. brucei* Service, *An. aruni* Sobti and *An. confusus* Evans and Leeson, and are difficult to distinguish morphologically (Gillies and Coetzee, 1987; Gillies and De Meillon, 1968). Despite the morphological similarities that exist between them, these species show ecological and behavioural differences that affect their vectorial capacities. *Anopheles funestus* is

*Corresponding author. Tel.: +27-11-489-9391; fax: +27-11-489-9399.
E-mail address: manti.weeto@nhls.ac.za (M.M. Weeto).

0035-9203/$ – see front matter © 2003 Royal Society of Tropical Medicine and Hygiene. Published by Elsevier Ltd. All rights reserved.
doi:10.1016/S0035-9203(03)00019-1

the only member of the group that is recognised as an important malaria vector because of its close association with humans and their habitations. *Anopheles rivulorum* is the only other member of the group that has been implicated as a minor vector of malaria in Tanzania (Wilkes et al., 1996). *Anopheles vaneedeni* was experimentally infected with *Plasmodium falciparum* in the laboratory, but has never been implicated in transmission in nature, although it feeds readily on humans outdoors (De Meillon et al., 1977). *Anopheles funestus* occurs in most parts of West, Central, East and southern Africa, Madagascar and the Comoros Islands (Mouchet et al., 1998).

Correct identification and knowledge of the distribution of vector and non-vector species is required before effective control efforts can be undertaken. For example Hargreaves et al. (2000) reported that although 4 members of the *An. funestus* group could be found resting inside houses in northern Kwazulu-Natal, South Africa, only the nominal species, *An. funestus*, was infected with malaria parasites.

Identification of members of the *An. funestus* group has, until recently, depended on the use of morphological characters (Gillies and Coetzee, 1987; Gillies and De Meillon, 1968) and polytene chromosomes (Green, 1982; Green and Hunt, 1980). Morphologically it is difficult to identify individual members of the group using adult characteristics, although immature features can distinguish some of the species. *Anopheles funestus*, *An. parensis*, *An. vaneedeni* and *An. aruni* form the *funestus* sub-group, being identical in almost all respects (Gillies and Coetzee, 1987). *Anopheles leesoni* is the most distinct member of the group with distinguishing characters at both the egg and larval stages. *An. rivulorum/brucei* and *An. confusus* can be identified on larval characteristics, although the former pair cannot be distinguished from each other. *Anopheles fuscivenosus* is known only from the adult stage and is similar to *An. leesoni* (Gillies and De Meillon, 1968). Morphological identification is by no means simple—egg batches from wild-caught blood-fed females need to be obtained and eggs, fourth instar larvae and adults examined per iso-female line. The rearing process takes 4—6 weeks and is extremely difficult. Cytogenetics, although more rapid than morphological examination, can only be used to identify half-gravid females of *An. parensis*, *An. rivulorum*, *An. leesoni*, *An. fuscivenosus* and *An. confusus* (Green, 1982). *Anopheles vaneedeni* can be distinguished from *An. funestus* using a polymorphic inversion on Arm 2; however if the inversion is homozygous for the standard arrangement, then these 2 species cannot be separated (Green and Hunt, 1980).

Molecular techniques have only recently been used for identifying members of the *An. funestus* group. They have the advantage of being applicable to any life stage and sex of the mosquito. The development of the single strand conformation polymorphism polymerase chain reaction (SSCP-PCR) assay made it possible to identify 4 members of the group: *An. rivulorum*, *An. funestus*, *An. vaneedeni* and *An. leesoni* (Koekemoer et al., 1999). *Anopheles parensis* was not included in the initial study and subsequent analysis of this species using SSCP yielded profiles indistinguishable from *An. vaneedeni*. Subsequently, an internal transcribed spacer PCR (ITS2-PCR) assay was developed that can distinguish between *An. parensis* and *An. vaneedeni* (Koekemoer et al., 2002a). By combining these two assays it is possible to identify all five species but the process becomes cumbersome and expensive. In order to differentiate these member species in a single reaction, a cocktail PCR assay was recently developed (Koekemoer et al., 2002b).

This paper evaluates the species-specific PCR assay using field specimens collected from a broad geographical range.

2. Materials and Methods

2.1. Collection of mosquitoes

Samples of field specimens collected from 14 localities were used (Table 1). Collection methods included house resting, pit traps and landing catches. The specimens were preserved on desiccated silica gel and transported to the Vector Control Reference Unit at the National Institute for Communicable Diseases (NICD), Johannesburg, South Africa. Morphological identification of the wild adults was carried out as described by Gillies and De Meillon (1968) and Gillies and Coetzee (1987) and those identified as belonging to the *An. funestus* group were assayed by PCR based techniques.

2.2. A priori Identification

DNA for all the samples was extracted from 2 legs or wings of each specimen as described by Collins et al. (1987).

All samples were amplified using the SSCP-PCR assay as described by Hiss et al. (1994) with minor alterations (Koekemoer et al., 1999). PCR amplification was carried out in a Hybaid thermal cycler under the following conditions: denaturing at 94 °C for 30 s, annealing at 40 °C for 30 s, extension at

Table 1 Specimens used in this study: collection and DNA extraction methods

Country	Locality	Collection method
Angola	Omnhenene (17°23′S, 14°34′E)	Resting indoors
Côte d' Ivoire	Petionara (7°42′N, 5°0′W)	Resting indoors
Ethiopia	Jimma (7°39′N, 36°47′E)	Cattle habitations close to human dwellings
Kenya	Mwea (37°35′E, 00°67′S)	Resting indoors
Madagascar	Anakazobe (18°19′S, 47°07′E)	Biting humans indoors
Malawi	Karonga (9°57′S, 33°55′E)	Resting indoors
Mozambique	Boane (26°02′S, 32°16′E)	Resting indoors
Namibia	Caprivi Strip (17°30′S, 24°16′E)	Resting outdoors
South Africa	Ndumu (27°02′S, 32°19′E)	Resting indoors
	Mamfene (27°23′S, 32°16′E)	Feeding on cattle
	Thomo (23°15′S, 30°47′E)	Biting humans indoors
Tanzania	Bagamoyo (6°26′S, 38°55′E)	Resting indoors
Uganda	Tororo (0° 42′N,34°12′E)	Resting indoors
Zambia	Chabona (14°24′S,28°28′E)	Spray catch

72 °C for 30 s for 30 cycles and a final extension at 72 °C for 10 min. 10 μl of the PCR product was visualised on an ethidium bromide-stained 2.5% agarose gel to check for successful amplification. The PCR products were denatured at 94 °C for 10 min and then put directly on ice. 10 μl of the sample was loaded onto the SSCP gel. Gels were run for 16—18 h at a constant temperature of 15 °C and a constant voltage of 200 V. Gels were then stained using the silver staining method of Budowle et al. (1991).

Those specimens tested by the SSCP electrophoresis method that gave *An. vaneedeni* banding patterns were re-amplified using the ITS2-PCR as described by Koekemoer et al. (2002a). Cycling conditions were set at 94 °C for 30 s, 50 °C for 30 s and 72 °C for 30 s for 40 cycles with a final extension step at 72 °C for 5 min. The amplified fragments were loaded on an ethidium bromide-stained 2.5% agarose gel and visualised under ultraviolet light.

2.3. Species-specific PCR

Amplification of the species-specific fragments was done according to Koekemoer et al. (2002b). Amplification cycles were as follows: denaturing at 94 °C for 30 s, annealing at 45 °C for 30 s and extension at 72 °C for 30 s, for 35 cycles followed by a final extension at 72 °C for 10 min. 10 μl of the amplified PCR product of each sample was loaded on a 2.5% ethidium bromide-stained agarose gel.

2.4. Controls

Each gel of each assay contained a DNA size standard, positive controls using morphologically identified species of *An. vaneedeni*, *An. rivulorum*, *An. parensis* and *An. leesoni* as well as *An. funestus* colony material as template DNA and a negative control.

3. Results and discussion

Of the 900 specimens identified by both the SSCP-PCR and ITS2-PCR assays, 868 (96.4%) were identified to the same species by the species-specific PCR (Table 2). The remaining 32 (3.5%) specimens all originated from Malawi and did not produce any amplification, despite being tested more than once and the positive controls producing results each time. The failure of the DNA of these samples to amplify cannot be ascribed to DNA degradation as SSCP-PCR examination of these samples was done after the species-specific PCR and an identification of *An. funestus* obtained. The inability of the species-specific PCR to identify these specimens, might indicate failure of primer annealing due to sequence variation. It is interesting that all 32 specimens that gave negative products are from the same locality, suggesting the possibility that an additional taxon within the *An. funestus* group may be present there. However, no sequencing of these individuals has been done to confirm this. The failure to identify these samples was not significantly different from the samples from the other countries (Chi-square; $P > 0.05$). For all the samples tested, the negative controls were always negative. Fig. 1 shows a typical species-specific PCR assay result that unambiguously identified all specimens.

A major disadvantage of using the SSCP-PCR and the ITS2-PCR assay as routine diagnostic tools is that they must be run concurrently for accurate species identification. The SSCP-PCR assay

Table 2 Summary of specimens collected in the wild and identified by SSCP, ITS2-PCR and the species-specific PCR

Samples (*n*)	Identification	Identification method		
		PCR-SSCP	ITS2-PCR	Species-specific PCR
Angola, Omnehene (12)	*An. funestus*	12	—	12
Cote d' Iviore, Ivory cast (21)	*An. funestus*	19	—	19
	An. leesoni	2	—	2
Ethiopia, Jimma (26)	*An. parensis*	26[a]		26
Kenya, Mwea (169)	*An. parensis*	169[a]	169	169
Madagascar, Anakazobe (2)	An. *funestus*	2	—	2
Malawi, Karonga (138)	*An. funestus*	138	—	106
Mozambique, Boane (57)	*An. funestus*	57	—	57
Namibia, Caprivi Strip (2)	*An. rivulorum*	2	—	2
South Africa, Mamfene (118)	*An. funestus*	26	—	26
	An. leesoni	1	—	1
	An. parensis	91[a]	55	55
	An. vaneedeni		36	36
South Africa, Ndumo (198)	*An. funestus*	13	—	13
	An. leesoni	92	—	92
	An. vaneedeni	82[a]	82	82
	An. rivulorum	11	—	11
South Africa, Thomo (51)	*An. funestus*	7	—	7
	An. leesoni	4	—	4
	An. vaneedeni	32[a]	32	32
	An. rivulorum	8	—	8
Tanzania, Bagamoyo (66)	*An. funestus*	56	—	56
	An. rivulorum	10	—	10
Uganda, Masese (22)	*An. funestus*	22	—	22
Zambia, Chabona village (18)	*An. funestus*	8	—	8
	An. leesoni	10	—	10

[a] Species that gave the same profile for *An. parensis* and *An. vaneedeni* on the SSCP-PCR were identified to the correct species using the ITS2-PCR.

can only reliably identify 3 species namely, *An. funestus*, *An. rivulorum* and *An. leesoni* and cannot distinguish between *An. vaneedeni* and *An. parensis* (Koekemoer et al., 1999). Samples that are identified as *An. vaneedeni* on the PCR-SSCP assay require a subsequent PCR based on the ITS2 region to differentiate between *An. vaneedeni* and *An. parensis*. (Koekemoer et al., 2002a). The

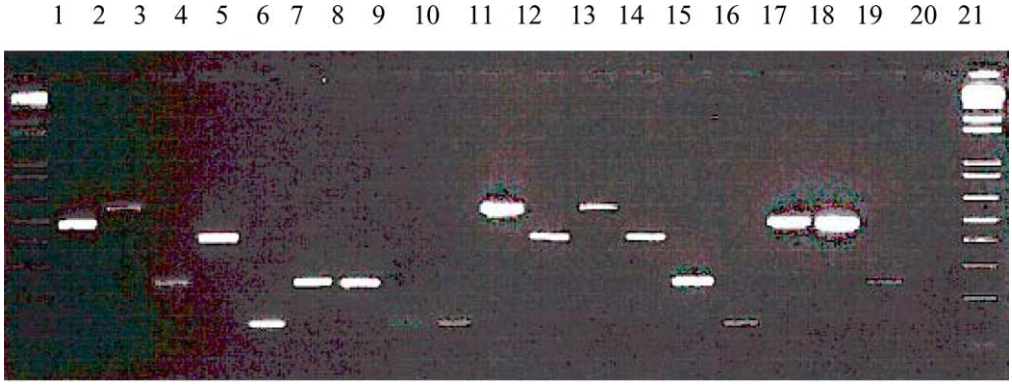


Fig. 1 A 2.5% agarose gel containing ethidium bromide showing fragments amplified by the species-specific polymerase chain reaction assay to identify members of the *An. funestus* group. Lane 1 and 21: 1 Kb-plus DNA ladder; lane 2—6: (positive controls), *An. funestus*, *An. vaneedeni*, *An. parensis*, *An. rivulorum*, *An. leesoni*; lane 7,8: *An. parensis* (Ethiopia), lane 15: *An. parensis* (Kenya), lane 19: *An. parensis* (South Africa, Mamfene); lane 9,10,16: *An. leesoni* (South Africa Thomo, Zambia, Cote d' Ivoire); lane 11 and 13: *An. vaneedeni* (South Africa Ndumo, Mamfene); lane 12 and 14: *An. rivulorum* (Namibia, Tanzania); lane 17 and 18: *An. funestus* (Angola, Zambia); lane 20: negative control.

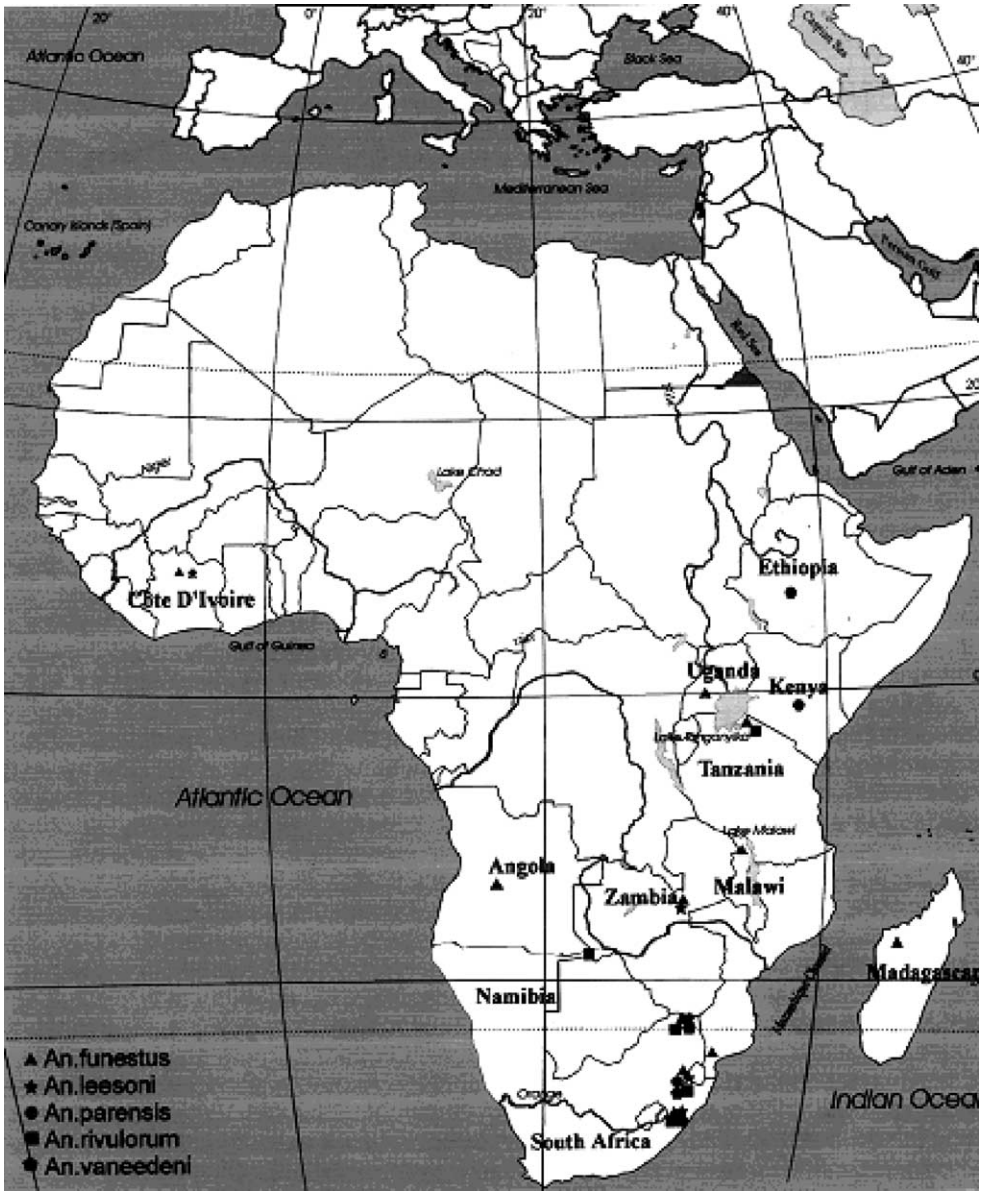


Fig. 2 Map of Africa and Madagascar showing members of the *An. funestus* group collected at various sites.

species-specific PCR eliminates this problem by distinguishing all 5 species in a single reaction. Furthermore, an average of 2 days is required to process DNA using the SSCP/ITS2-PCR method. This time is reduced by 50% using the new technique. No specialised equipment or solutions are required. Laboratories in Africa already using PCR to identify members of the *An. gambiae* complex can easily include this new assay without additional expenses.

The study shows high numbers of *An. funestus* from most of the collection sites, and in some cases occurring in sympatry with other members of the group (Fig. 2). Thus, not only is the new technique more convenient in terms of cost effectiveness and speed, but also is essential for areas where members of the *An. funestus* group may occur in sympatry.

Acknowledgements

The following people are thanked for collection of field specimens: B. Ntomwa (Angola, Namibia); B. Brooke (Cote d'Ivoire, Mozambique); M. Fettene (Ethiopia); J. Chiphwanya (Malawi); K. Hargreaves, T. Furumele (South Africa); T. Byembabazi (Uganda); V. Siachinji (Zambia). This study was supported in part by the NHLS Research Foundation, the SA Medical Research Council, The Wellcome

Trust, and the World Health Organization, Regional Office for Africa.

References

Budowle, B., Chakraborty, R., Giusti, A.M., Eisenberg, A.J., Allen, R.C., 1991. Analysis of the VNTR locus DIS80 by the PCR followed by high resolution PAGE. Am. J. Hum. Genet. 48, 137—144.

Collins, F.H., Mendez, M.A., Rasmussn, P.C., Mahaffey, H.J., Besansky, N.J., Finnerty, V., 1987. A ribosomal RNA gene probe differentiates member species of the *Anopheles gambiae* complex. Am. J. Trop. Med. Hyg. 37, 37—41.

De Meillon, B., Van Eeden, G.J., Coetzee, L., Coetzee, M., Meiswinkel, R., Du Toit, L.N., Hansford, C.F., 1977. Observations on a species of the *Anopheles funestus* subgroup, a suspected exophilic vector of malaria parasites in northeastern Transvaal, South Africa. Mosquito News 37, 657—661.

Gillies, M.T., Coetzee, M., 1987. A supplement to the Anophelinae of Africa south of the Sahara. Publ. S. Afr. Inst. Med. Res, no. 55.

Gillies, M.T., De Meillon, B., 1968. The Anophelinae of Africa South of the Sahara. Publ. S. Afr. Inst. Med. Res. no. 54.

Green, C.A., 1982. A cladistic analysis of mosquito chromosome data (*Anopheles* (Cellia) Myzomyia). J. Hered. 73, 2—11.

Green, C.A., Hunt, R.H., 1980. Interpretation of variation in ovarian polytene chromosomes of *Anopheles funestus* Giles, *A. parensis* Gillies and *A. aruni*? Genetica 51, 187—195.

Hargreaves, K., Koekemoer, L.L., Brooke, B.D., Hunt, R.H., Mthembu, J., Coetzee, M., 2000. *Anopheles funestus* resistant to pyrethroid insecticides in South Africa. Med. Vet. Entomol. 14, 181—189.

Hiss, R.H., Norris, D.E., Dietrich, C.H., Whitcomb, R.F., West, D.F., Bosio, C.F., Kambhampati, S., Piesman, J., Antolin, M.F., Black, W.C., Hogh, B., 1994. Molecular taxonomy using single-strand conformation polymorphism (SSCP) analysis of mitochondrial ribosomal DNA genes. Insect Mol. Biol. 3, 171—182.

Koekemoer, L.L., Lochouran, L., Hunt, R.H., Coetzee, M., 1999. Single-strand conformation polymorphism analysis for identification of four members of the *Anopheles funestus* (Diptera: Culicidae) group. J. Med. Entomol. 36, 125—130.

Koekemoer, L.L., Hargreaves, K., Hunt, R.H., Coetzee, M., 2002a. Identification of *Anopheles parensis* (Diptera: Culicidae) using ribosomal DNA internal transcribed spacer (ITS2) sequence variation. Afr. Entomol. 10, 235—239.

Koekemoer, L.L., Kamau, L., Hunt, R.H., Coetzee, M., 2002b. A cocktail polymerase chain reaction assay to identify members of the *Anopheles funestus* (Diptera: Culicidae) group. Am. J. Trop. Med. Hyg. 66, 804—811.

Mouchet, J., Manguin, S., Sircoulon, J., Laventure, S., Faye, O., Onapa, A.W., Carnevale, P., Julvez, J., Fontenille, D., 1998. Evolution of malaria in Africa for the past 40 years: impact of climatic and human factors. J. Am. Mosq. Control Assoc. 14, 121—130.

Wilkes, T.J., Matola, Y.G., Charlwood, J.D., 1996. *Anopheles rivulorum*, a vector of human malaria in Africa. Med. Vet. Entomol. 10, 108—110.

Available online at www.sciencedirect.com

SCIENCE DIRECT®